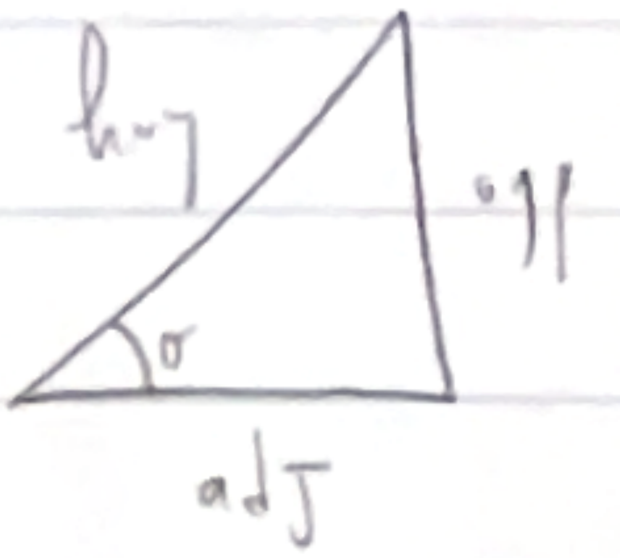
W2

Trigonometric functions

→ in periodic phenomena ← acoustic waves
water waves
circular movement



$$\sin\theta = \frac{opp}{hyp}$$

$$\cos\theta = \frac{adj}{hyp}$$

$$\tan\theta = \frac{opp}{adj} = \frac{\sin\theta}{\cos\theta}$$

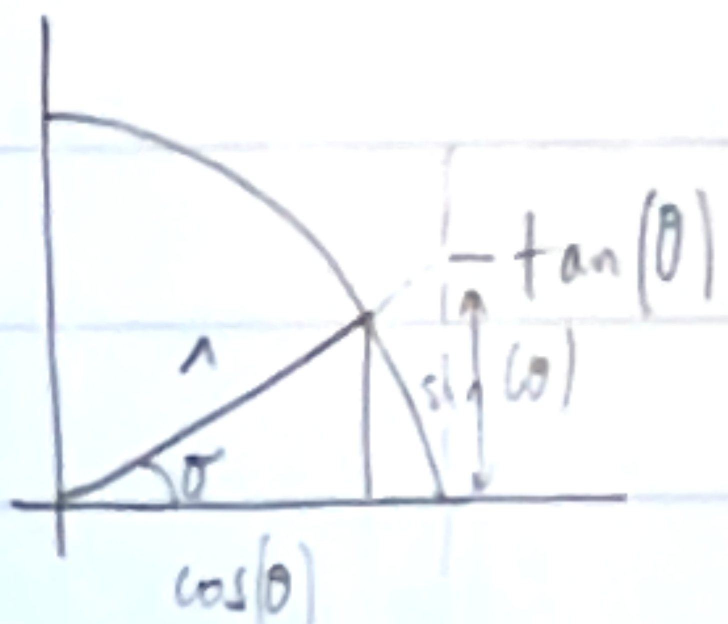
radiants

$2\pi \text{ rad} = 360°$

$1 \text{ rad} = \left(\frac{180}{\pi}\right)°$

$$s = \frac{\theta}{2\pi} \cdot 2\pi r = r \cdot \theta$$

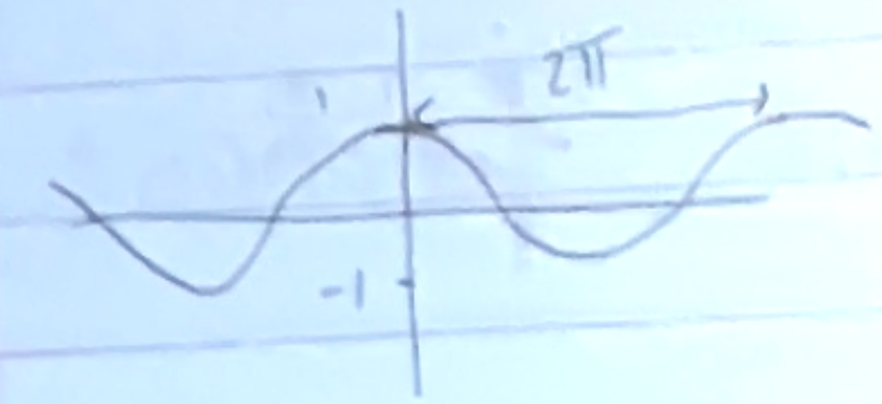
$1° = \frac{\pi}{180} \text{ rad}$



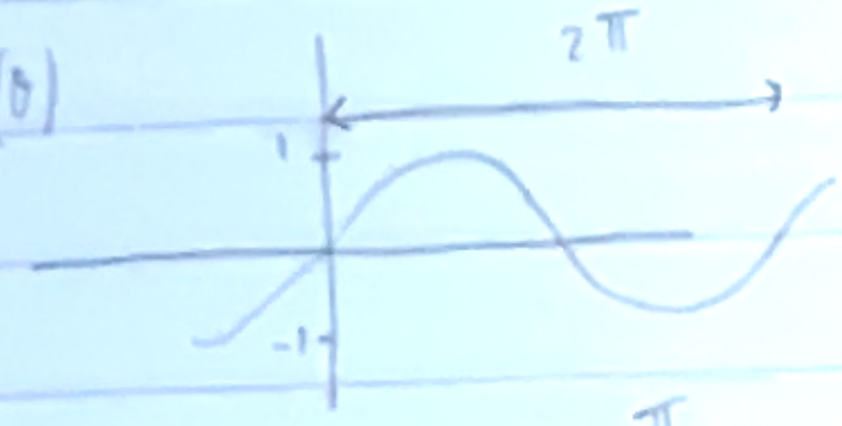
for special angles

	0°	30°	45°	60°	90°
θ	0	$\frac{1}{6}\pi$	$\frac{1}{4}\pi$	$\frac{1}{3}\pi$	$\frac{1}{2}\pi$
$\sin(\theta)$	0	$\frac{1}{2}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}\sqrt{3}$	1
$\cos(\theta)$	1	$\frac{1}{2}\sqrt{3}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}$	0
$\tan(\theta)$	0	$\frac{1}{3}\sqrt{3}$	1	$\sqrt{3}$	—

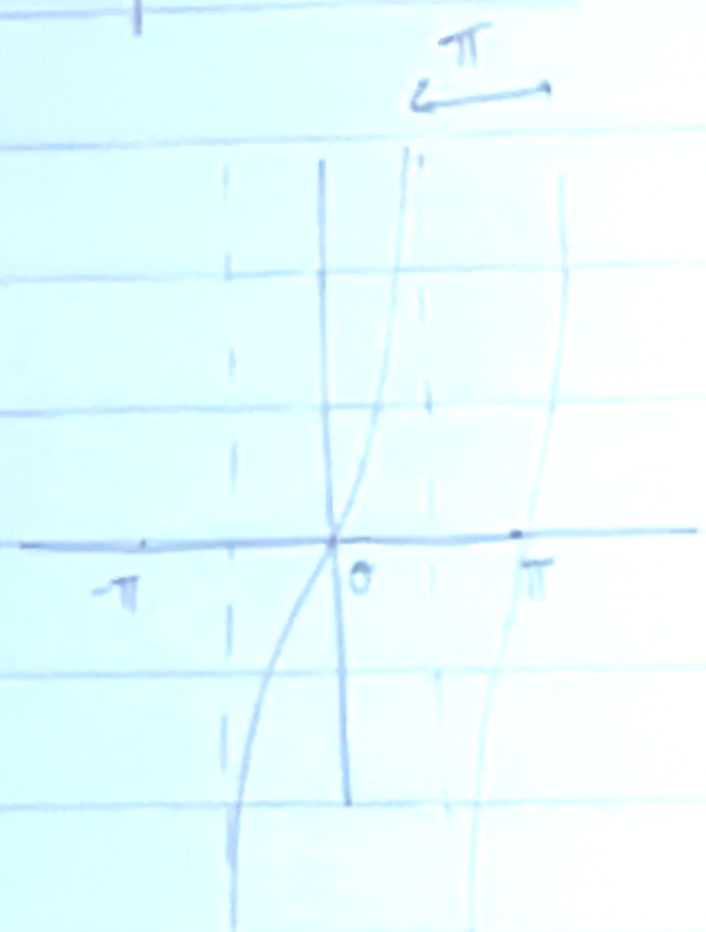
$\cos(\theta)$



$\sin(\theta)$



$tg(\theta)$



Rules of Calculation

Pythagorean Theorem

$$\cos^2(\theta) + \sin^2(\theta) = 1$$

Periodicity

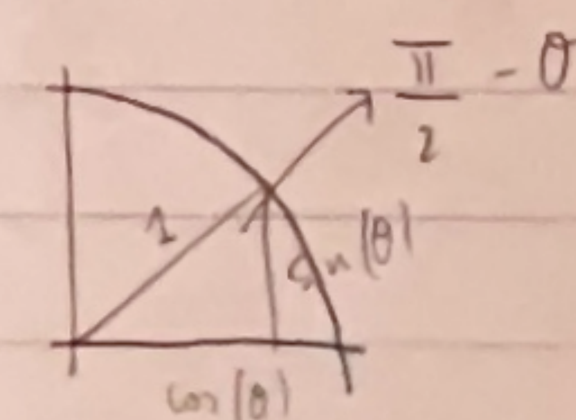
$$\cos(\theta + 2\pi) = \cos(\theta) \qquad \text{every } 2\pi$$
$$\sin(\theta + 2\pi) = \sin(\theta)$$

$$\cos(\theta - 2\pi) = \cos(\theta)$$
$$\sin(\theta - 2\pi) = \sin(\theta)$$

$tg(x)$ periodic → but in this case period π

$$\tan(x \pm \pi) = \tan(x)$$

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin(\theta)$$



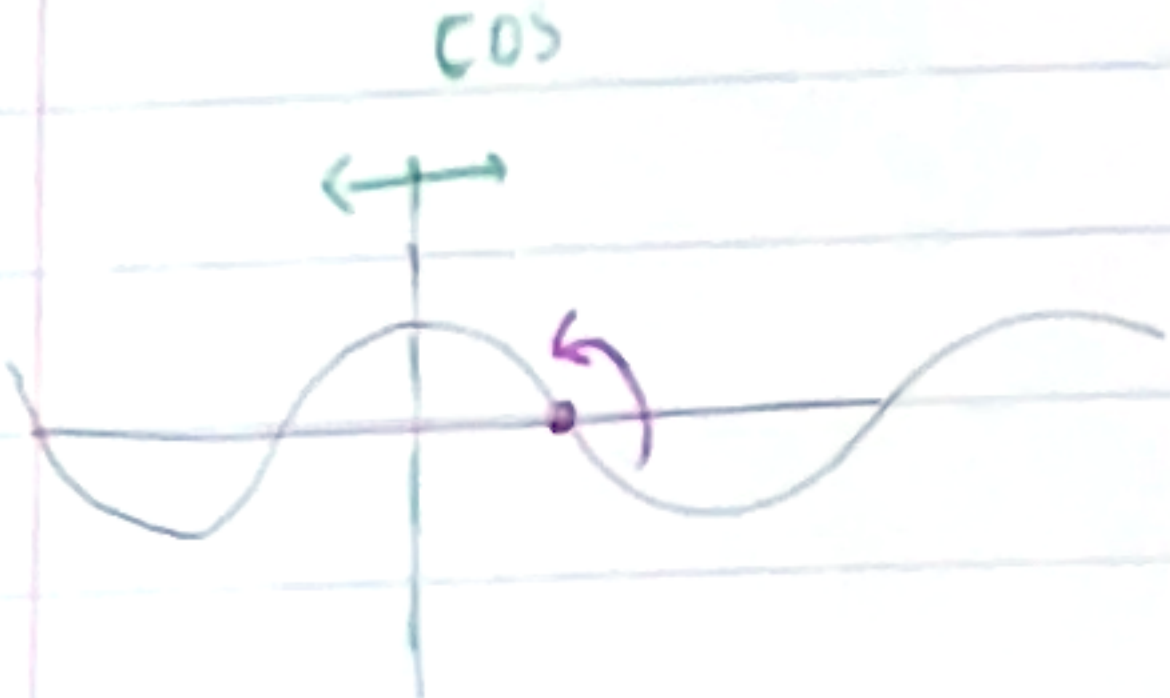
$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta$$

$\cos(\theta)$ even function:

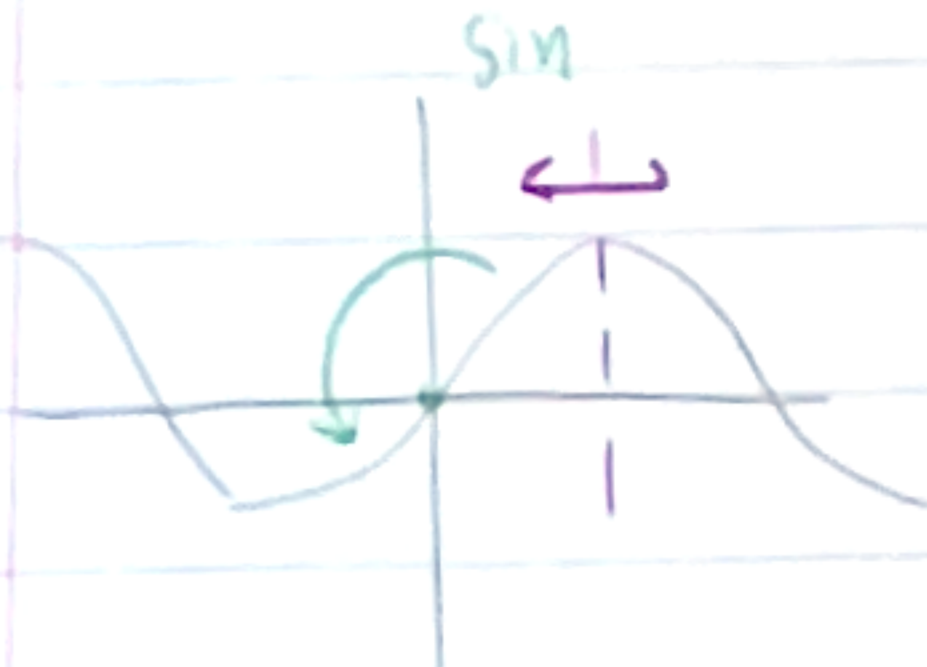
- $\cos(\theta) = \cos(-\theta)$

Origin — Sine is odd $f(x)$.

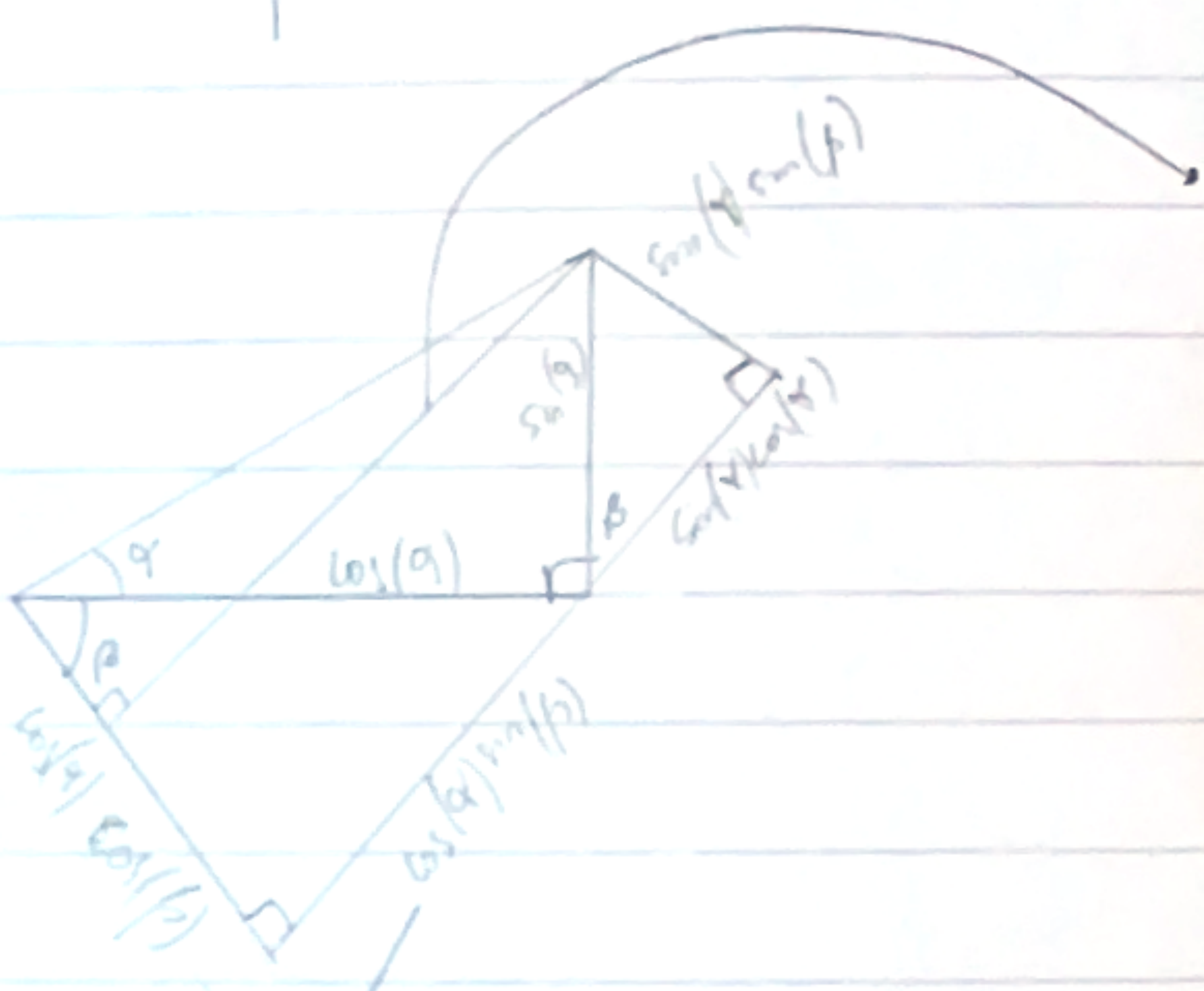
- $\sin(-\theta) = -\sin(\theta)$



- $\cos(\pi - x) = -\cos(x)$



- $\sin(\pi - x) = \sin(x)$



as hypotenuse is $\cos(\alpha)$

$\sin(\alpha + \beta) =$
$\cos(\alpha)\sin(\beta) + \sin(\alpha)\cos(\beta)$

$\cos(\alpha + \beta) =$
$\cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)$

$\sin(2x) = 2\sin(x)\cos(x)$

$\cos(2x) = \cos^2(x) - \sin^2(x)$

↓

this proof for acute angles $\alpha + \beta$

↓

equations are for all $\alpha + \beta$

7.3 Compositions

→ order matters

$f(g(x))$

inner function

outer function

Translation → gráficas se mueven _

Scaling

$f(g(x)) = 2g(x)$ → twice the value of $g(x)$ → distance to x-axis is multiplied by 2

$g(x) + a$	vertical shift upwards
$g(x+a)$	horizontal shift to the left
$ag(x)$	vertical scaling by a
$g(ax)$	horizontal scaling by $1/a$